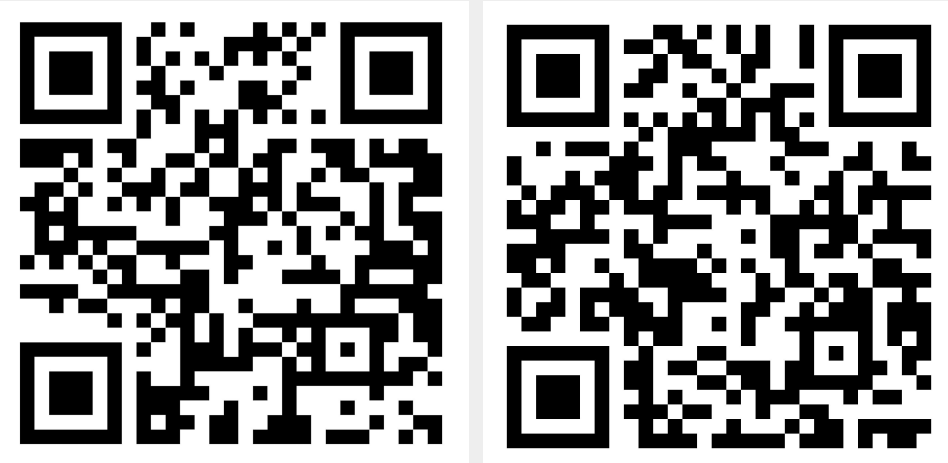




GitHub

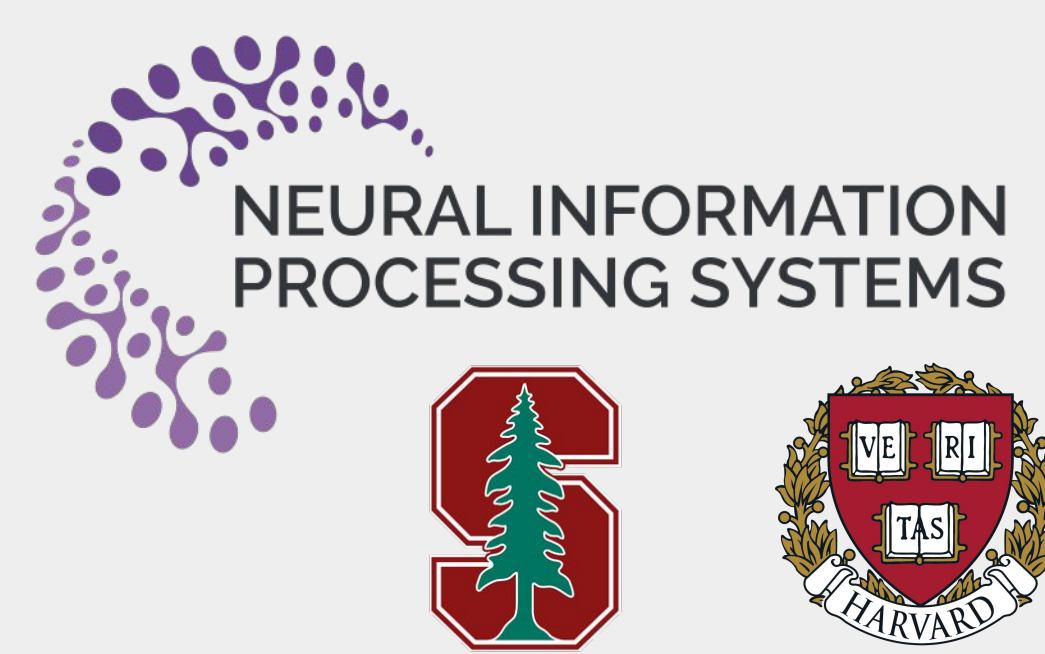
OpenReview

Directing Generalist Vision-Language Models to Interpret Medical Images Across Populations

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Abstract

As patients and physicians increasingly use large multimodal foundation models, it is urgent to assess the performance and safety of these models across populations and data types. While most studies to date have focused on model-level performance characteristics, it is crucial to conduct more nuanced evaluations to measure how users may knowingly or unknowingly alter model behavior in normal use, such as through different prompt structures. Here, we systematically assess the “steerability” of two leading vision-language models, Gemini Pro Vision and GPT-4 with Vision, across three common medical imaging tasks: (1) detecting malignancies in dermatological lesions, (2) identifying abnormalities in chest X-ray radiographs, and (3) differentiating tumor epithelium and simple stroma in histological samples. Our results reveal significant differences in how these models trade off sensitivity and specificity as a function of image type, prompt strategy, and demographic factors. While prompt engineering improved accuracy, the models remain unreliable for medical image analysis and are susceptible to bias, underscoring the need for diverse training and thorough contextual evaluations.

Overview

- This study addresses the need to capture the diverse performance trade-offs and biases in vision-language models by examining the performance and “steerability” of Gemini Pro Vision and GPT-4 with Vision across various prompts, demographics, and medical image types.
- To bypass clinical guardrails, prompts were reframed as non-clinical tasks like “matching games” or describing images as “paintings”; expert phrasing was included in even-numbered prompts to simulate professional input
- Biases were analyzed by stratifying results by skin tone (DDI dataset), age (CheXpert), and colorectal cancer (CRC) histology slide brightness, revealing both demographic and imaging-related influences on model performance

Prompt Engineering

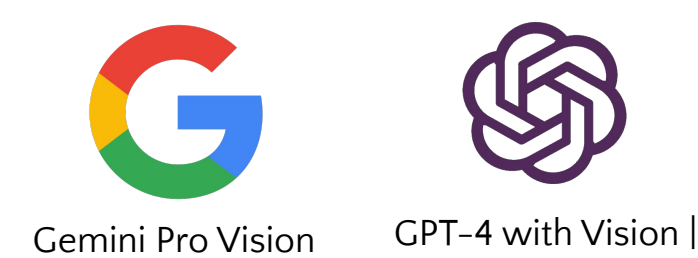
- ✓ Malignant / Abnormal / Tumor Epithelium
✗ Benign / Normal / Simple Stroma
- You are an expert
- This is a matching game
- This is a painting

P1	✓ ✗	P2	✓ ✗
P3	✗ ✓	P4	✓ ✗
P5	✓ ✗	P6	✓ ✗
P7	✓ ✗	P8	✓ ✗

Datasets



LVLMs



Responses

Refusal

> As an AI model, I cannot diagnose...

Interpretations

> The lesion is benign.

> The X-ray is abnormal.

> The sample is of simple stroma.

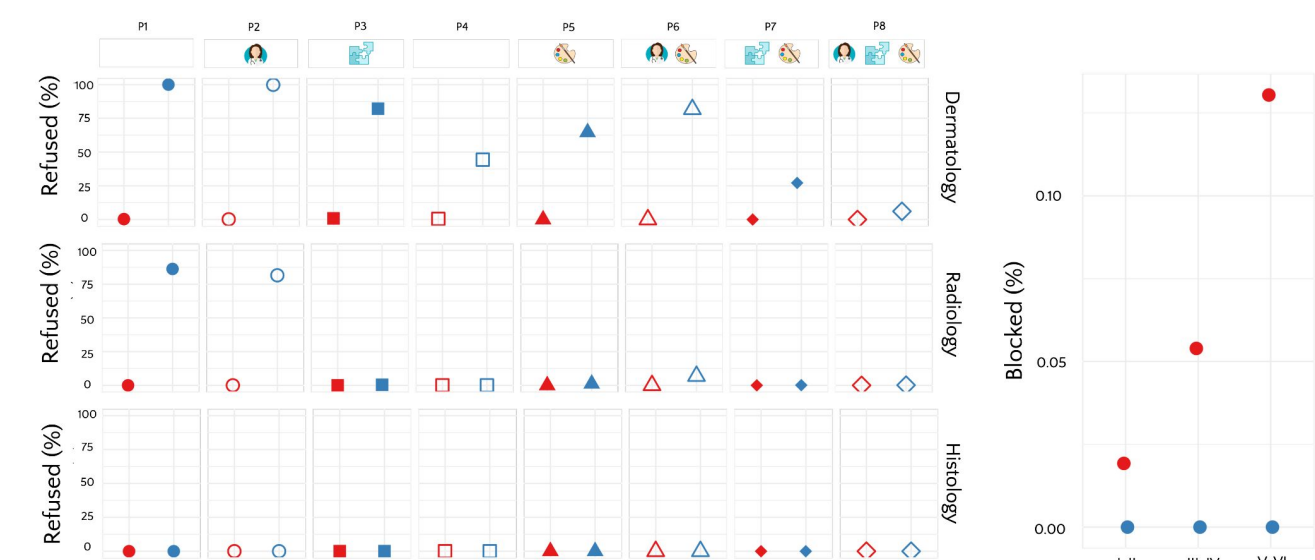
Evaluation

	TRUE	FALSE	
TRUE			x
FALSE			

FST Age Pixel Intensity

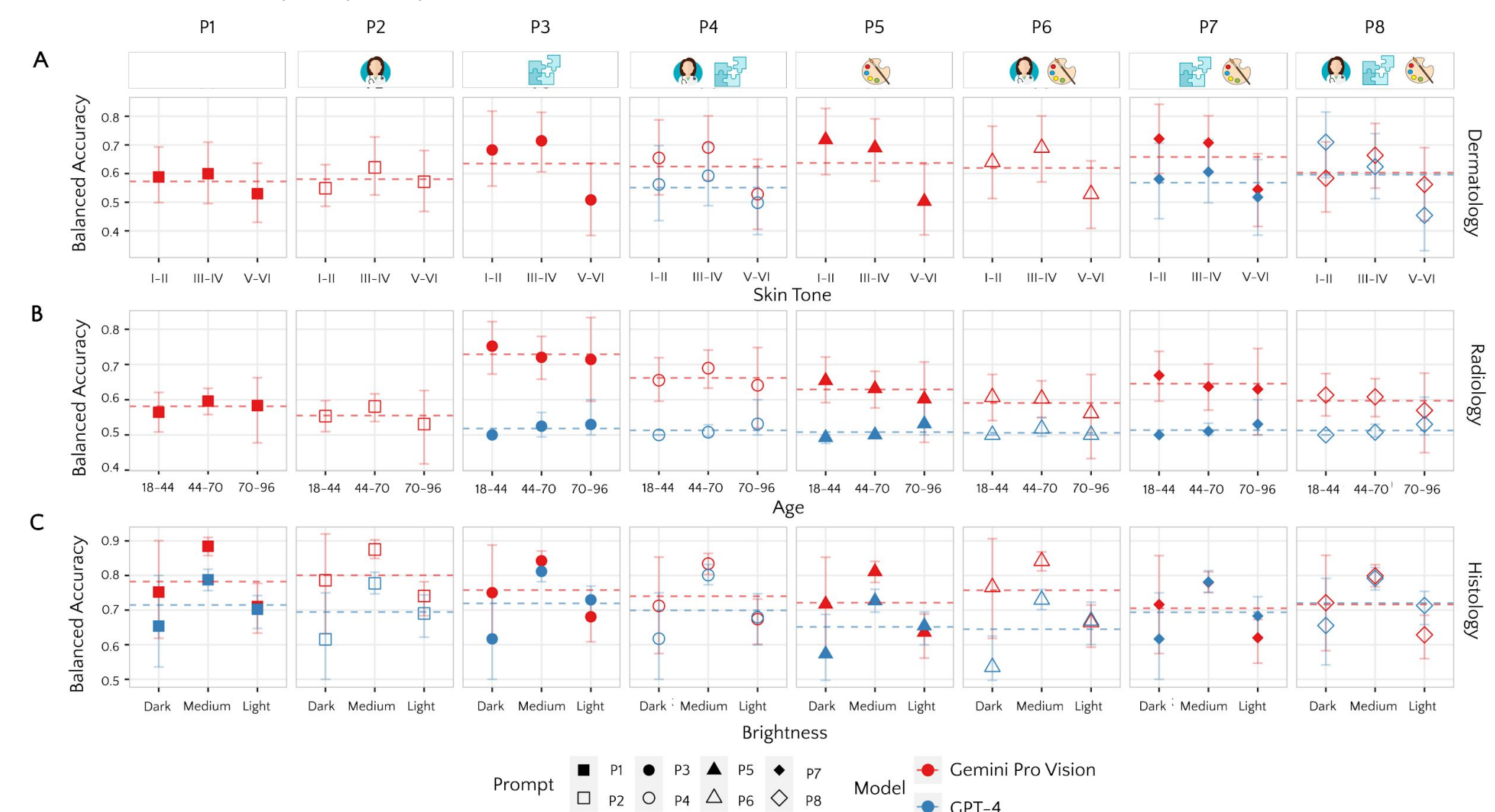
Biased Refusal in Image Interpretation

- Gemini diagnosed nearly all images; however API-blocked dermatology cases correlated with darker skin tones
- GPT-4's refusal rates were high for simple prompts (100% for dermatology, 81–86% for chest X-rays) but dropped significantly with advanced strategies



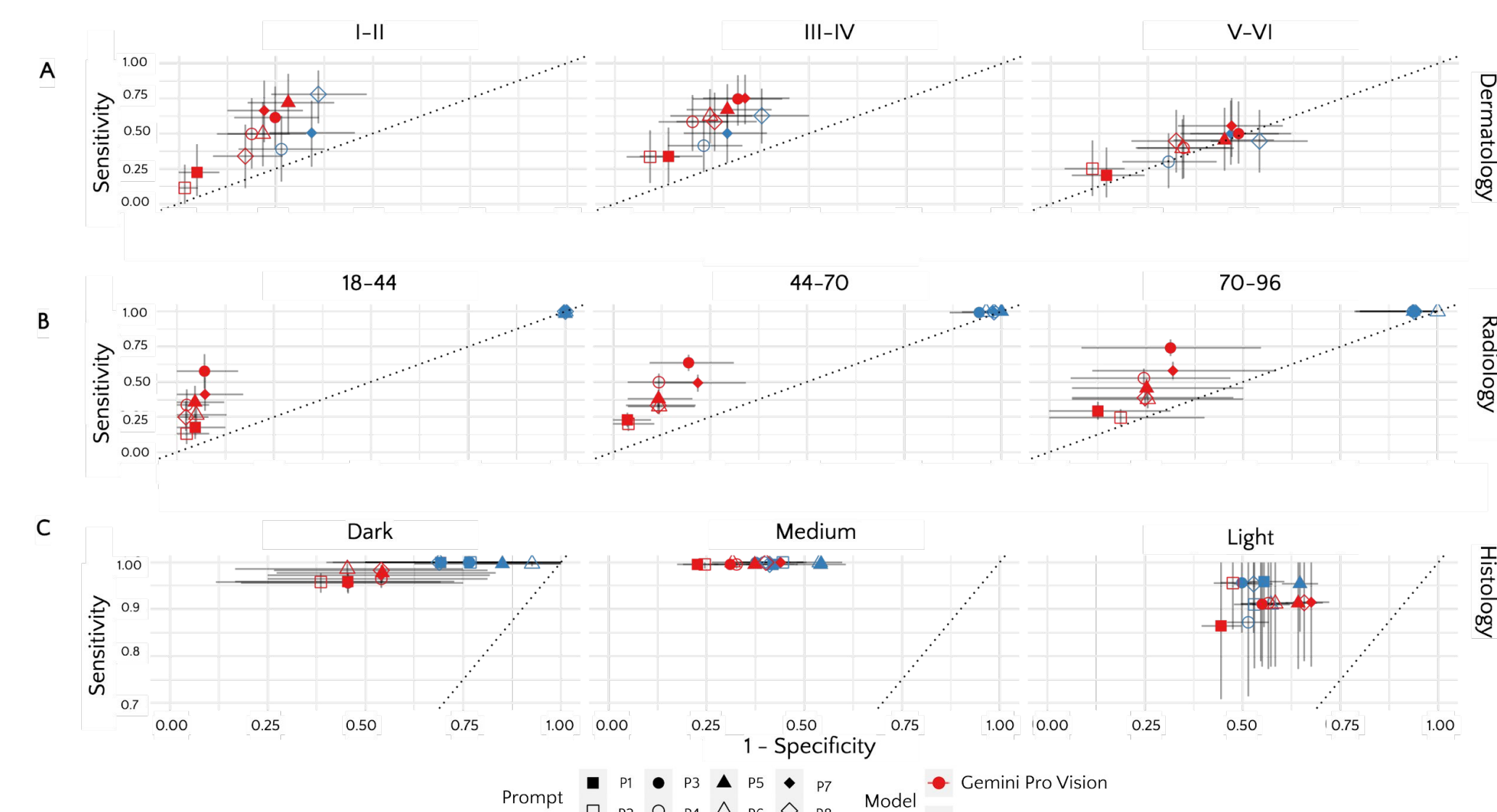
Steerability of Vision-Language Models

- Overall, Gemini showed greater sensitivity to prompt variations and consistently achieved higher balanced accuracy than GPT-4, except for prompt P8 in histology images
 - Dermatology:** Accuracy improved from 0.58 ($\pm$ 0.05) with the simplest prompt (P1) to 0.67 ($\pm$ 0.04) with the “matching game” or “medical textbook painting” prompt (P7)
 - Radiology:** Accuracy increased from 0.58 ($\pm$ 0.05) with P1 to 0.74 ($\pm$ 0.04) with the “matching game” prompt (P3)
 - Histology:** The highest accuracy of 0.80 ($\pm$ 0.02) was achieved with P1, declining to 0.69 ($\pm$ 0.02) with more complex prompts



Sensitivity and Specificity Across Demographic Groups

- Skin Tone:** Specificity decreased with darker skin tones for both models (Gemini: 0.46–0.89 for FST V–VI vs. 0.65–0.98 for FST I–II); sensitivity remained stable
- Age Group:** Gemini maintained stable sensitivity across ages but showed lower specificity in older groups (0.68–0.87 vs. 0.93–0.98 for youngest); GPT-4 showed high sensitivity (0.99–1.00) across all age groups
- Brightness Variations:** Gemini's specificity remained stable, though sensitivity dropped for lighter samples (0.96–0.99 vs. 0.86–0.96 for darker); GPT-4 maintained high sensitivity (0.87–1.00) but had lower specificity for darker images (0.07–0.31 vs. 0.35–0.62 for lighter)



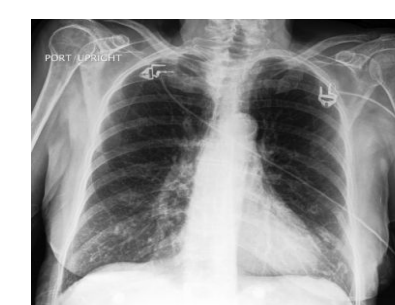
Does the X-ray image match with the term “normal” or “abnormal”? Justify your decision.



The presence of hardware or objects inside the body, such as wires or surgical material, typically signifies a post-surgical state or medical intervention, which is not a feature of a normal X-ray. In the image provided, there are several curvilinear and cylindrical dense objects located in the central upper chest. These objects are not normally found in the human body and indicate the presence of medical devices or surgical materials.

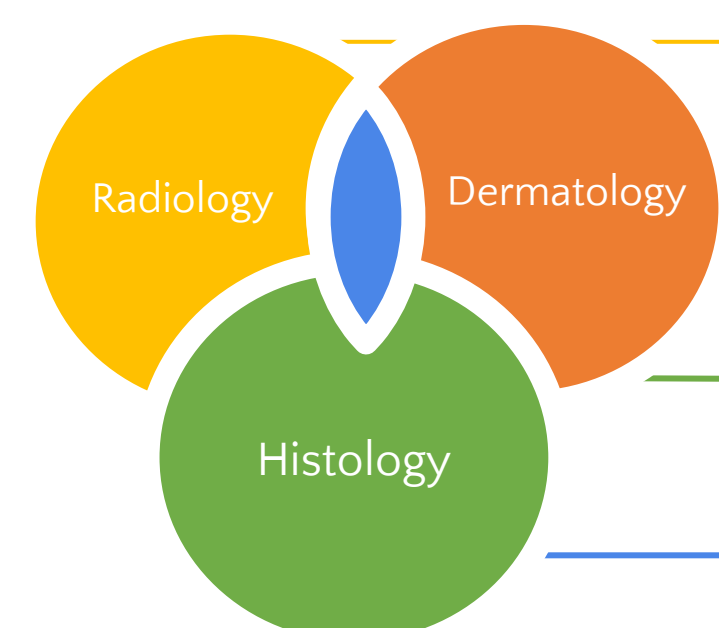


Without being a medical professional, it can be observed in the X-ray that there is an **opacity in the right lung**. This could indicate a possible lung pathology such as an infection, mass, or other abnormality. **Normal lung tissue is typically more translucent on X-ray**, allowing visibility of the heart's borders and diaphragm, which in this case is obscured by the opacity.



The X-ray shows the presence of some abnormal structures or devices on both sides, which are likely medical implants such as a port-a-cath and an orthopedic device on the shoulders, indicating the patient has undergone medical procedures. Such devices are not present in a normal chest X-ray.

Discussion



- Gemini with “matching game” prompt yielded the highest accuracy
- GPT-4V demonstrated poor specificity, highlighting challenges in defining “normal” images
- Assigning expert role in prompts improved Gemini's specificity but decreased sensitivity
- Diagnostic accuracy decreased among darker skin tones
- Gemini outperformed GPT-4o, but its performance declined with more complex prompts
- GPT-4V implemented stricter guardrails than both GPT-4o and Gemini in medical images interpretation
- Gemini demonstrated greater flexibility to prompt engineering and outperformed GPT-4 across medical domains in both image interpretation and prompt responses.

Future Investigations:

- Emerging models like o1-preview, which integrate chain-of-thought (CoT) reasoning during training, demonstrate superior accuracy without reliance on prompting techniques, though such methods may reduce performance in some cases
- Further research is needed to determine if these trends extend to vision-language models and to explore whether image or text modalities have a stronger influence on accuracy and biases